

CO 4 - ASSESSMENT TOOL -2

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Subject Code : CSA1003
Subject Name : Software Engineering

Hotel Reservation and Room Management Platform with Dynamic Pricing

CI/CD Pipeline Design Exercise

1. Project Overview

The **Hotel Reservation and Room Management Platform with Dynamic Pricing** is a full-stack application that provides online hotel room reservation, room availability management, customer management, administrator operations, and dynamic room pricing.

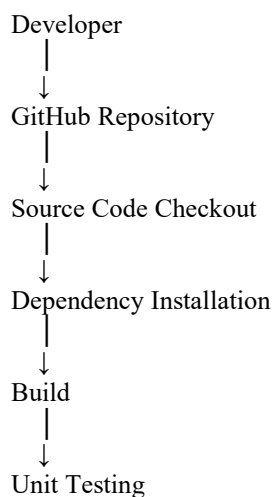
Customers can register, log in, search for rooms, view room details, check availability, and make reservations. Hotel administrators can manage rooms, reservations, customers, pricing rules, and reports.

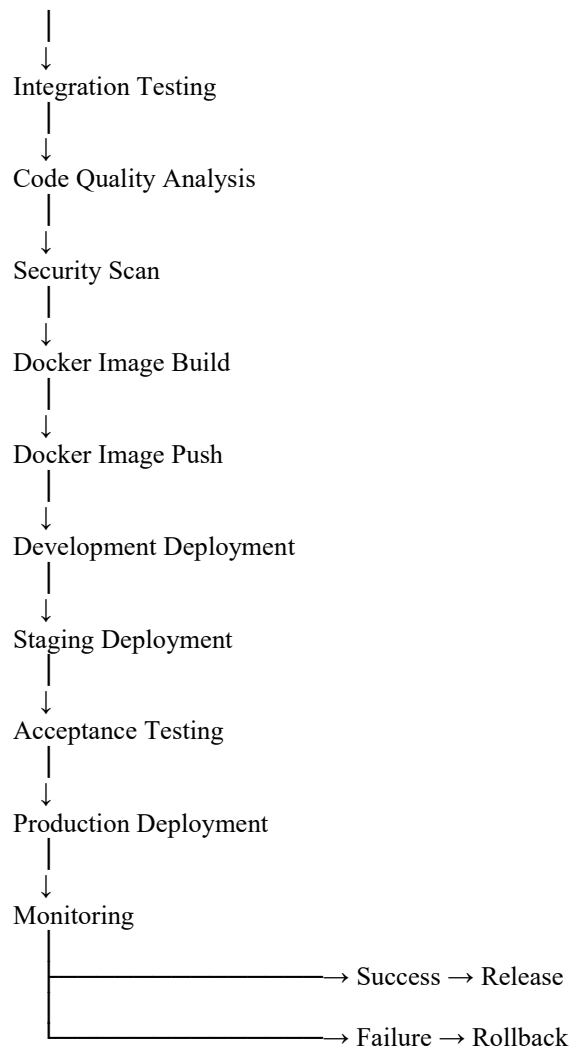
A CI/CD pipeline is required to automatically build, test, analyze, package, and deploy the application whenever developers make changes to the source code.

The assignment specifically requires a pipeline covering source code management, build, testing, quality analysis, packaging, deployment, security, notifications, and rollback.

3. Proposed CI/CD Pipeline

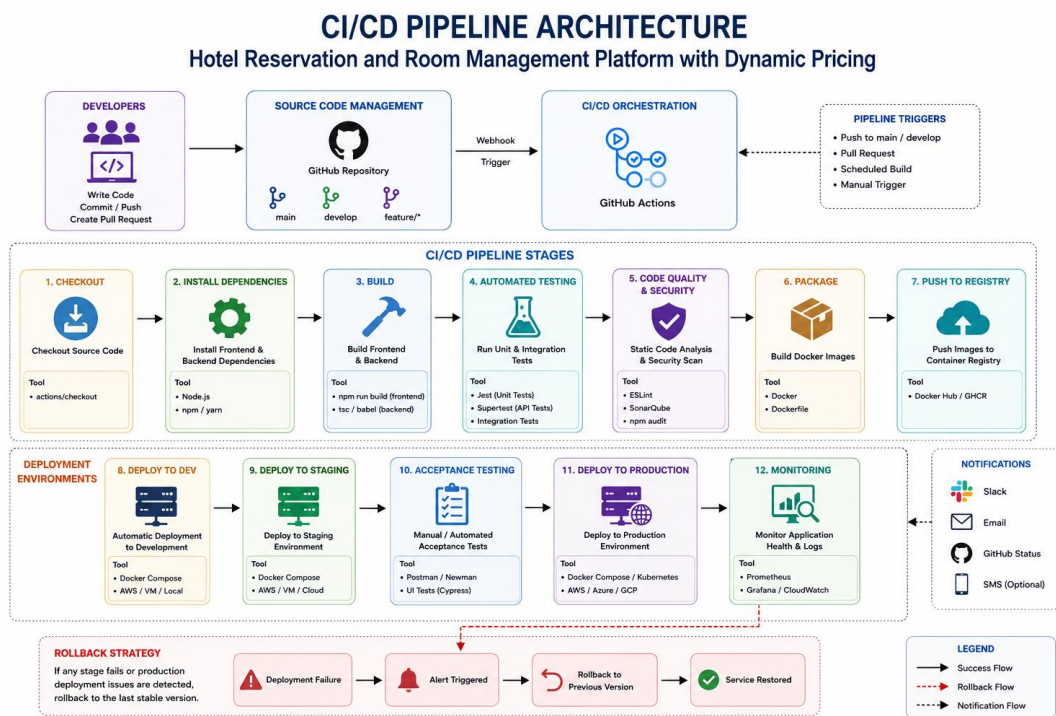
The proposed pipeline is:





This structure covers the major pipeline stages expected by the assessment rubric: source, build, testing, quality checks, packaging, and deployment.

4. CI/CD Pipeline Architecture



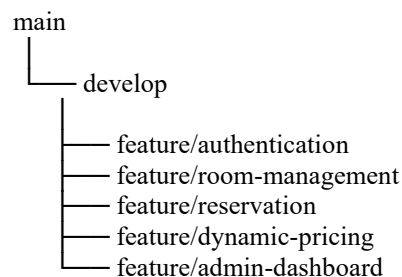
5. Tools and Technology Selection

Pipeline Stage	Tool/Technology	Purpose
Source Control	Git	Version control
Repository	GitHub	Source code hosting
CI/CD	GitHub Actions	Pipeline automation
Frontend	React.js	User interface
Backend	Node.js + Express.js	REST APIs and business logic
Database	MySQL	Persistent data storage
Testing	Jest	Automated unit testing

6. Source Code Management

The project uses Git and GitHub for source code management.

Branch Structure



7. Build Stage

Frontend

```
cd frontend  
npm install  
npm run build
```

Backend

```
cd backend  
npm install
```

8. Automated Testing Strategy

Unit Testing

Integration Testing

API Testing

9. Dynamic Pricing Testing

Since dynamic pricing is a core feature, it should receive dedicated automated tests.

Low Demand

Normal Demand

High Demand

High Occupancy

11. Security Checks

Dependency Scanning

Environment Variables

12. Docker Packaging

After successful testing and quality checks, Docker images are created.

Frontend Image

```
docker build -t hotel-frontend ./frontend
```

Backend Image

```
docker build -t hotel-backend ./backend
```

The images can then be pushed to Docker Hub:

```
docker push hotel-frontend
docker push hotel-backend
```

13. Deployment Environments

Production

Production is the environment used by actual customers and hotel administrators.

```
Approval
↓
Production Deployment
↓
Monitoring
```

Only successfully tested builds should reach production.

The assignment specifically requires development, testing/staging, and production environments in the deployment strategy.

14. Deployment Strategy

A controlled deployment strategy is used.

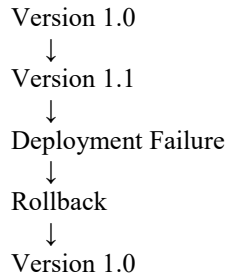
```
Code Commit
↓
Automated CI
↓
All Tests Pass?
/\
No  Yes
|   |
Stop ↓
    Staging
    ↓
    Acceptance Test
    ↓
    Approval
    ↓
    Production
```

Production deployment should only occur after all required tests and quality checks pass.

15. Rollback Strategy

If a production deployment fails, the system should return to the previous stable version.

Example:



16. Monitoring and Notifications

After deployment, the application should be monitored for:

Application failures.

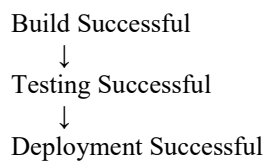
API errors.

Database connectivity.

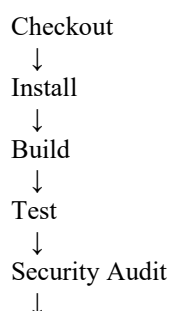
Container health.

Notifications

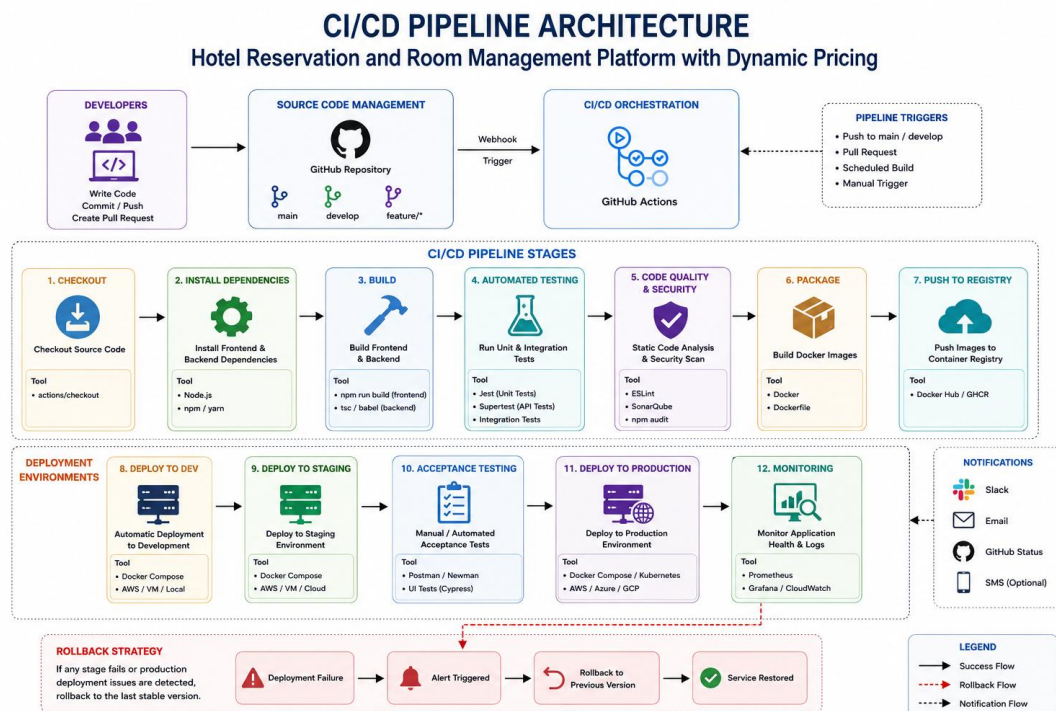
The CI/CD pipeline can notify developers when:



17. Example GitHub Actions Workflow



18. Complete CI/CD Workflow



19. Advantages of the Proposed CI/CD Pipeline

Faster Development

Early Error Detection

Improved Quality

Reliable Deployment

Better Security

Reduced Downtime

Continuous Delivery

Better Collaboration

20. Future Enhancements

The CI/CD pipeline can be further improved by adding:

Kubernetes deployment.

Cloud-based infrastructure.

Blue-green deployment.

Canary deployment.

Automated database migrations.

Advanced vulnerability scanning.

21. Conclusion

The proposed CI/CD pipeline for the Hotel Reservation and Room Management Platform with Dynamic Pricing automates the complete software delivery lifecycle from code commit to production deployment. The pipeline integrates GitHub, GitHub Actions, automated testing, ESLint, security scanning, Docker, container registry, staging, production deployment, monitoring, and rollback. The design addresses the major evaluation areas specified in the assignment: requirement and pipeline analysis, pipeline architecture, tool selection, automated testing and quality integration, deployment/security/rollback, and pipeline documentation.